

Overview of the OpenROAD Digital Design Flow from RTL to GDS

Sherief Reda

Brown University

In this talk, we review the current status of OpenROAD, which is an open-source end-to-end silicon compiler. The goal of OpenROAD is to reduce cost, expertise, schedule and risk barriers that face system designers today. The open-source flow builds around an industrial strength open-source design database, and incorporates a number of design tools that include physical synthesis synthesis, floorplanning, placement, clock tree synthesis, global routing and detailed routing. The flow also incorporates analysis and support tools for static timing analysis, parasitic extraction, power integrity analysis, and cloud deployment. The entire flow can be customized through scripting and enables users to experiment with various tool options and optimizations.